



SILVER OAK UNIVERSITY

College of Technology

Bachelor of Technology

Computer Engineering

Course Name: Mobile Application Development

Course Code:1010043368

Semester:6th

Prerequisite:

Basic Knowledge of OOPS concept and Core java

Objective:

1. Understand the fundamentals of Android OS, its architecture, UI components, and activity/fragment lifecycles.
2. Learn Dart programming and develop cross-platform applications with Flutter using widgets, layouts, and themes.
3. Apply Android and Flutter concepts to build scalable, user-friendly mobile applications for industrial projects.

Teaching Scheme:

Teaching Scheme				
L	T	P	Contact Hours	Credit
1	0	2	3	2

Contents:

Unit	Topics	Teaching Hours	Weightage %
1	Unit:1 Android OS Introduction to Android, Android Architecture, Building Blocks of Android Application, Work with Activity, Activity Lifecycle, Intents Fragments, Fragment Lifecycle	3	20
2	Unit:2 Android UI Component Create Android UI, Work with Layouts, Create Custom Layouts, Work with UI Components and Events, Material Design Toolbar, Tab Layout, Recycler View and Card View, Android Menus	3	25
3	Unit:3 Database Connectivity and Applicability to Industrial Projects Storage in Android, Shared Preferences, Android Requesting Permission at run time (Android 6.0), JSON Parsing, Using Third Party Library: Retrofit	3	20
4	Unit:4 Fundamentals of Dart Programming for Flutter Working with Data Types, Variables and Constants, Functions, Loops, Control Flow, Classes and Objects.	2	15
5	Unit:5 Working with UI in Flutter	4	20

	Widgets, Containers and Rows/Columns, Text and Fonts, Images, Buttons, Forms and Inputs Fields, Navigations, Dialogs and Icons, Styles and Themes.		
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Course Outcomes:

Sr. No.	CO Statement	Unit
CO-1	Create an application using the basic fundamentals of Android.	1
CO-2	Develop Android Application using UI Component	2
CO-3	Implement fully functional Mobile application with database and Third party library in Android.	3
CO-4	Demonstrate the use of DART programming.	4
CO-5	Develop Mobile Application using UI Component in Flutter.	5

Teaching & Learning Methodology:

The various methods or tools to teach the above course:

1. PPT
2. Video Lectures etc.

List of Experiments:

Total Hours: 28

Sr. No.	Practical Name
1	Android studio installation and Create a “Hello World” application. That will display “Hello World” in the middle of the screen using the Text View Widget in the red color.
2	Design Registration activity using necessary UI Components: Implement using the following layouts: a. Linear Layout b. Relative Layout c. Constraint Layout
3	Create Activities & implement the following: a. Implicit intent b. Explicit Intent c. StartActivityForResult
4	Create Android applications that demonstrate the Android activity life cycle and concept of fragments.
5	Create Android applications that demonstrate the concept of the menu with navigation.
6	Create an Android app to display student details in Listview (using Adapter class).
7	Create an Android application that stores user Login credentials using Shared Preferences.
8	To implement parsing JSON data using the Retrofit library in Android application.
9	Write a program in Dart that finds simple interest. Formula= $(p * t * r) / 100$.
10	Create an application using commonly used Widgets in Flutter (Text, padding, margin, and alignment).
11	Create a stateless widget and use it in your main application.
12	Create a stateful widget with a mutable state.
13	Create forms for user input with validation.
14	Display images in your app using the Image widget.

Books Recommended:

1. Android Application Development Black Book by Pradeep Kothari, DreamTech
2. Beginning Android 4 Application Development by Wei Meng Lee, Wrox
3. Android Wireless Application Development By Lauren Darcey, Shane Conder, Pearson
4. Flutter Cookbook By Simone Alessandria, Brian Kayfitz.

List of Open Source Software/learning website:

1. <http://silveroakuni.ac.in/video-lecture>
2. <https://nptel.ac.in/>
3. <http://www.coursera.org/>
4. <https://developer.android.com/>
5. Latest configured Computer systems
6. Android Studio Software

CO-PO-PSO Matrix:

Co. No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO-1	3	2	2		2								1	3	2
CO-2	2	2	3	2	3						2	2	2	2	2
CO-3	3	3	3	3	3							2	2	3	2
CO-4	2	2	2	2	3						2	2	3	3	2
CO-5	2	2	3	2	3						2	3	2	3	2